

Tensor computations

Generic identifiability of tensors at strictly subcritical ranks

Difficulty: extreme **Importance:** interesting to the community

Statement

Let $d \geq 3$, $n_1 \geq \dots \geq n_d \geq 2$, and define

$$r_* = \frac{\prod_{i=1}^d n_i}{1 + \sum_{i=1}^d (n_i - 1)}, \quad b = \prod_{i=2}^d n_i - \sum_{i=2}^d (n_i - 1).$$

For every integer $1 \leq r < r_*$, a generic rank- r tensor in $\bigotimes_{i=1}^d \mathbb{C}^{n_i}$ is conjectured to have a unique expression as a sum of r nonzero rank-one tensors, except in these cases:

1. $n_1 > b$ and $r \geq b$.
2. Format $(4, 4, 3)$, rank 5.
3. Format $(n, n, 2, 2)$, rank $2n - 1$, with $n \geq 2$.
4. Format $(4, 4, 4)$, rank 6.
5. Format $(6, 6, 3)$, rank 8.
6. Format $(2, 2, 2, 2, 2)$, rank 5.

A rank-one summand is $v_1 \otimes \dots \otimes v_d$. Uniqueness identifies decompositions whose tensor summands differ only by permutation. Generic means outside a proper Zariski-closed subset of the Zariski closure of tensors of rank at most r , restricted to tensors of actual rank r . The number r_* is a rational dimension-count value, not an assumed actual generic rank.

Relevance

The conjecture specifies the range in which a generic low-rank tensor model has uniquely recoverable factors, an essential precondition for numerical CP decomposition and component interpretation.

References

1. L. Chiantini, G. Ottaviani, and N. Vannieuwenhoven, *Effective criteria for specific identifiability of tensors and forms*. [Primary preprint](#), §3, Conjecture 6 and Remark 7, p.7. The author journal manuscript numbers these Conjecture 3.4 and Remark 3.5.
2. The same authors, *An Algorithm for Generic and Low-Rank Specific Identifiability of Complex Tensors*, SIAM J. Matrix Anal. Appl. 35 (2014), 1265–1287. [DOI](#); [primary preprint](#), Theorem 1.1, for the verified bounded-format classification.
3. A. Massarenti and M. Mella, *Bronowski's conjecture and the identifiability of projective varieties*. [January 2024 primary revision](#), abstract and §§3–4, for later conditional links between identifiability and nondefectivity.

Status check — 2026-09-08

Reference 1 records verification for $\prod n_i \leq 15000$. The later Massarenti–Mella results reduce broad identifiability questions to secant nondefectivity under hypotheses; they are not a full classification of these formats. Searches "generic identifiability" "conjecture" "2026" tensor, "Chiantini" "Vannieuwenhoven" "conjecture" "2025", and "Chiantini" "Vannieuwenhoven" "conjecture" "2026" located no full resolution. Status is bounded by those searches.